

# Browser Privacy Agent — Feature List

2026-09-22 · @Someone

90 features in six lanes, one owner per lane. S  $\approx$  half a person-week, M  $\approx$  one person-week; Week is the target weekend merge.

## Lane A — Platform

| ID   | Feature               | Done when                                                                                                                        | Size | Week | Deps       | Status      |
|------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------|------|------|------------|-------------|
| A-01 | Monorepo and builds   | Vite + @crxjs Chrome build, web-ext Firefox build, CI running lint and tests on every PR                                         | S    | 1    | —          | Not started |
| A-02 | Platform adapter      | platform/chrome.ts and platform/firefox.ts behind one interface; webextension-polyfill wired                                     | S    | 1    | A-01       | Not started |
| A-03 | Typed message bus     | Request/response with timeouts between background, content script, offscreen doc and worker, using protocol types; stub handlers | M    | 2    | A-01       | Not started |
| A-04 | ML host lifecycle     | Chrome offscreen document and Firefox background host start the worker; kept alive during a task, released after 10 min idle     | M    | 3    | A-03       | Not started |
| A-06 | Network chokepoint    | One fetch client in background calling the Egress Guard hook; ESLint rule fails builds that call fetch /XHR elsewhere            | S    | 4    | A-03       | Not started |
| A-08 | Coordinate transforms | Viewport, page, image, crop and frame-local conversions with DPR, zoom and scroll; property-based tests                          | S    | 4    | —          | Not started |
| A-12 | Session client        | Start, step, feedback and escalate calls through the chokepoint; SSE stream parsing; network retries reuse the same seq          | M    | 5    | A-06, E-02 | Not started |
| A-11 | Invocation manager    | commands hotkey and push-to-talk hotkey; mode state (Ask, Guide, Agent, Background) shared across contexts                       | S    | 6    | A-03       | Not started |

| ID   | Feature                | Done when                                                                                                                                 | Size | Week | Deps       | Status      |
|------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------|------|------------|-------------|
| A-13 | Model store            | CDN download, SHA-256 check against pinned hashes, IndexedDB cache, lazy-load API used by lane C                                          | S    | 6    | A-04       | Not started |
| A-10 | Side panel shell       | Chrome side panel and Firefox sidebar hosting F-01 components: task input, action list, stop button, settings route                       | M    | 7    | A-02       | Not started |
| A-05 | Permission manager     | activeTab flow; runtime <code>optional_host_permissions</code> requests per origin with site-named prompt; revoke from settings           | M    | 8    | A-02       | Not started |
| A-07 | Screenshot capture     | <code>captureVisibleTab</code> with 2/s limiter; data URL to offscreen; decode to <code>ImageBitmap</code> ; zero-copy transfer to worker | M    | 9    | A-04       | Not started |
| A-09 | Frame manager          | <code>all_frames</code> injection, frame offset chain, <code>opaque_frame</code> elements when no host permission                         | M    | 10   | A-05, B-02 | Not started |
| A-14 | Metrics-only telemetry | Step latency, backend, counts; payloads impossible to log by type; user opt-out                                                           | S    | 11   | A-06       | Not started |
| A-15 | Release packaging      | Signed Chrome zip, Firefox AMO build, version bump and changelog script                                                                   | S    | 11   | A-01       | Not started |

## Lane B — DOM & Executor

| ID   | Feature                         | Done when                                                                                                                                                | Size | Week | Deps | Status      |
|------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|-------------|
| B-01 | Element model                   | Element, bbox, state, <code>src</code> , <code>doc_id</code> types in <code>packages/protocol</code> ; fixtures for tests                                | S    | 1    | E-01 | Not started |
| B-16 | Offset-preserving normalization | NFC, whitespace and case folding with an index map back to the original; grapheme clusters via <code>Intl.Segmenter</code> ; shared by NER and alignment | S    | 1    | —    | Not started |

| ID   | Feature                           | Done when                                                                                                                                                            | Size | Week | Deps       | Status      |
|------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------------|-------------|
| B-02 | DOM walker                        | <code>TreeWalker</code> over document, open shadow roots and same-origin iframes; candidate filter; runs in $\leq 8$ ms idle chunks                                  | M    | 2    | B-01       | Not started |
| B-04 | Visibility and occlusion          | Zero-size, hidden, clipped, opacity filters; 5-point <code>elementFromPoint</code> occlusion flag                                                                    | S    | 3    | B-02       | Not started |
| B-05 | Element registry and doc identity | <code>WeakMap</code> + <code>Map&lt;id, WeakRef&gt;</code> + <code>FinalizationRegistry</code> sweep; <code>doc_id</code> changes on navigation and SPA route change | S    | 3    | B-02       | Not started |
| B-08 | Mock sites pack 1                 | Deterministic local sites: profile edit form, settings page, search + filter list; served in CI                                                                      | M    | 4    | —          | Not started |
| B-10 | Click adapter                     | Buttons, links, checkbox, radio via focus + <code>element.click()</code> ; tested through the extension, not Playwright clicks                                       | S    | 5    | B-05       | Not started |
| B-11 | Text input adapter                | Native value setter + <code>input / change</code> ; <code>clear_first</code> ; works on React and Vue controlled inputs in mock sites                                | S    | 5    | B-05       | Not started |
| B-03 | Accessible name and role          | ARIA, <code>&lt;label for&gt;</code> , <code>aria-labelledby</code> , implicit roles, placeholder/alt fallbacks; 50-case fixture suite                               | M    | 6    | B-02       | Not started |
| B-06 | Incremental extraction            | <code>MutationObserver</code> dirty subtrees; step 2+ re-extracts only changed parts; diff output (added, changed, removed)                                          | M    | 7    | B-05       | Not started |
| B-07 | Settle detector                   | 300 ms DOM quiet, 3 s cap; optional page-world fetch/XHR counter with its limits documented                                                                          | S    | 8    | B-06       | Not started |
| B-14 | Verification engine               | Evaluates <code>expect ( value_equals</code> via vault compare hook, appear/disappear, URL change); returns ok, failed, mismatch or unknown                          | M    | 9    | B-07, B-11 | Not started |

| ID   | Feature                          | Done when                                                                                                                                            | Size | Week          | Deps | Status      |
|------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------|------|-------------|
| B-15 | Freshness and consequence checks | Rejects stale <code>doc_id / observation_id</code> ; target-drift tolerance; consequential-action classifier; blocks auto-replay of unknown outcomes | S    | 10            | B-14 | Not started |
| B-09 | Mock sites pack 2                | Multi-page form, repeated identical controls, canvas widget, SPA routing, shadow DOM, cross-origin iframe                                            | M    | 11            | B-08 | Not started |
| B-12 | Select, Enter and Tab adapters   | Native select, custom dropdown via clicks, Enter as <code>requestSubmit()</code> , Tab as focus move                                                 | M    | 12            | B-10 | Not started |
| B-13 | Rich editor and canvas adapters  | Contenteditable via <code>beforeinput / insertText</code> ; canvas pointer events; emits <code>unsupported_control</code> for Guide fallback         | M    | 13 (cut line) | B-10 | Not started |

## Lane C — Vision

| ID   | Feature                    | Done when                                                                                                                                       | Size | Week | Deps | Status      |
|------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|-------------|
| C-01 | Worker inference API       | <code>runDetector</code> , <code>runOCR</code> , <code>runFaces</code> , <code>runIcons</code> interface with fake outputs other lanes can call | S    | 1    | —    | Not started |
| C-02 | ONNX runtime               | Backend selection WebGPU → WASM threads → WASM; cross-origin isolation for threads; session cache; tensors released per step                    | M    | 2    | C-01 | Not started |
| C-03 | In-browser model benchmark | Page that measures latency and memory per model per backend on the reference laptop; produces the M1 table                                      | S    | 3    | C-02 | Not started |
| C-08 | Face detector              | BlazeFace-class model on image, video and canvas regions; boxes in page coordinates                                                             | S    | 3    | C-02 | Not started |

| ID   | Feature                     | Done when                                                                                                                                                   | Size | Week | Deps       | Status      |
|------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------------|-------------|
| C-04 | Detector dataset pipeline   | Headless crawl renders pages, converts DOM bboxes and roles to YOLO labels; split by site family                                                            | M    | 4    | —          | Not started |
| C-16 | Canary leak CI              | Proxy captures all egress; decompresses and decodes bodies; searches canaries defined by lane D; OCR + face checks on captured images; degraded-device runs | M    | 5    | A-06, B-08 | Not started |
| C-07 | OCR                         | PaddleOCR mobile det + rec in ONNX; crop batching; word boxes with per-character confidence; Latin + Devanagari                                             | M    | 6    | C-02       | Not started |
| C-05 | Detector v1                 | YOLO-nano trained, ONNX export, int8 quantization, mAP reported on held-out site families                                                                   | M    | 7    | C-04       | Not started |
| C-06 | Detector in worker          | 640 px letterbox, NMS, output converted image → page coordinates with A-08                                                                                  | S    | 8    | C-05, A-08 | Not started |
| C-13 | Script identification       | Small classifier on OCR line crops: Latin, Devanagari, other                                                                                                | S    | 8    | C-07       | Not started |
| C-10 | Interaction-map fusion      | IoU ≥ 0.5 matching of DOM and vision; <code>v*</code> vision-only elements with OCR names and confidence                                                    | M    | 9    | C-06, B-02 | Not started |
| C-11 | Tier selection              | Per-element Tier 0/1/2 decision: gap detection where DOM is empty but pixels show a control; conf < 0.6 flags Tier 2                                        | S    | 10   | C-10       | Not started |
| C-12 | QR and barcode locator      | Native <code>BarcodeDetector</code> where present, else ~1 MB WASM locator; returns boxes only, never decodes                                               | S    | 10   | C-02       | Not started |
| C-14 | Detector v2 privacy classes | Hand-labelled set adds avatar, photo, video, document_image, handwriting, map, chart; retrain and export                                                    | M    | 11   | C-05       | Not started |

| ID   | Feature                      | Done when                                                                                       | Size | Week          | Deps | Status      |
|------|------------------------------|-------------------------------------------------------------------------------------------------|------|---------------|------|-------------|
| C-09 | Icon classifier              | Synthetic dataset from icon fonts, MobileNetV3-small trained and exported, ~150 labels          | M    | 12            | C-02 | Not started |
| C-15 | Pixel diff and adaptive mode | 64×64 perceptual-hash tiles skip unchanged regions; device probe picks model set and reports it | S    | 13 (cut line) | C-06 | Not started |

## Lane D — Privacy

| ID   | Feature               | Done when                                                                                                                                       | Size | Week | Deps       | Status      |
|------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------------|-------------|
| D-01 | PII engine API        | Finding and redaction types; stub redactor that masks everything so early end-to-end runs are safe; canary case list for C-16                   | S    | 1    | E-01       | Not started |
| D-04 | Vault                 | Worker-memory store, consistent placeholders per session, cleared on task end, tab close or 15 min idle                                         | S    | 1    | D-01       | Not started |
| D-02 | Pattern rules (L2)    | Email, +91/E.164 phone, Luhn card, Verhoeff Aadhaar, PAN, IFSC, UPI, IBAN, OTP-in-context, JWT/API keys; 300+ fixtures                          | M    | 2    | D-01       | Not started |
| D-05 | Text redaction        | Placeholder substitution in names, values, text context; <code>value_state</code> ; semantic obfuscation keeps sentence meaning                 | M    | 3    | D-02, D-04 | Not started |
| D-07 | NER dataset and model | Faker-generated pages (Indian locale, Hindi) plus public PII corpora; small token classifier fine-tuned and exported; per-class recall reported | M    | 4    | —          | Not started |
| D-11 | Egress Guard          | Schema check, coverage check, field-aware vault matching, rule re-scan, block and surface reason                                                | M    | 5    | A-06, D-05 | Not started |

| ID   | Feature                                | Done when                                                                                                                                 | Size | Week | Deps                   | Status      |
|------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------|------|------------------------|-------------|
| D-08 | NER in worker                          | Transformers.js inference, token → character offsets through B-16; tiny fallback model validated to the same recall target                | M    | 6    | D-07, C-02, B-16       | Not started |
| D-03 | Semantic DOM (L1) and site policy (L5) | type / autocomplete /label keyword classes; per-site always-redact selectors, never-send list, user-marked regions                        | M    | 7    | B-03                   | Not started |
| D-09 | Vault use bindings                     | Task, origin, doc_id, allowed targets, operations enforced locally; binding_denied; confirm-destination hook for F-06                     | M    | 8    | D-04                   | Not started |
| D-10 | Restricted egress mode                 | Per-field detector coverage records; {{TEXT_WITHHELD}}; UI-label allowlist vocabulary; auto-enters when a detector is unavailable         | M    | 9    | D-08                   | Not started |
| D-12 | Span-to-pixel alignment                | DOM Range.getClientRects() and OCR word-box mapping; multiline; uncertain alignment masks the whole block                                 | M    | 10   | B-16, C-07             | Not started |
| D-13 | Crop redaction pipeline                | Opaque face fills, OCR → rules + NER masks, 4 px dilation, JPEG re-encode, region manifest                                                | M    | 11   | D-12, C-08             | Not started |
| D-14 | Crop allowlist enforcement             | Supported-content classes applied: barcodes and unsupported scripts masked, unclassified regions masked, manifest checked by Egress Guard | M    | 12   | D-13, C-12, C-13, C-14 | Not started |

## Lane E — Server

| ID   | Feature          | Done when                                                                                                             | Size | Week | Deps | Status      |
|------|------------------|-----------------------------------------------------------------------------------------------------------------------|------|------|------|-------------|
| E-01 | Protocol package | JSON Schemas for session, Screen State, Action, escalate, feedback; generated TS and Pydantic types; round-trip tests | M    | 1    | —    | Not started |

| ID   | Feature                 | Done when                                                                                                                     | Size | Week | Deps       | Status      |
|------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|------|------|------------|-------------|
| E-02 | Agent API skeleton      | FastAPI session endpoints, Redis store with 30 min TTL, scripted fake planner returning valid actions                         | M    | 2    | E-01       | Not started |
| E-03 | Planner serving         | vLLM on the GPU node with the M1-chosen 7B-class model, int8, prefix caching; OpenAI-compatible endpoint                      | M    | 3    | —          | Not started |
| E-05 | Prompt builder          | Cached system prefix with redaction contract; compact line format for elements; last-5-step memory                            | M    | 4    | E-02       | Not started |
| E-04 | Constrained decoding    | Guided decoding against the Action schema; 0 invalid outputs over 1,000 runs; latency benchmark                               | S    | 5    | E-03, E-01 | Not started |
| E-12 | Guide-mode outputs      | <code>point</code> , <code>annotate</code> , <code>say</code> schemas and prompt variant; <code>say</code> capped at 25 words | S    | 5    | E-05       | Not started |
| E-09 | Planner eval harness    | Runs the task set on mock sites; success rate, steps, p50/p95 latency; report per merge                                       | M    | 6    | B-08, E-05 | Not started |
| E-06 | Action validator        | Target exists, navigate allowlist, placeholder known; regenerate once, then <code>ask_user</code>                             | S    | 7    | E-05       | Not started |
| E-13 | Ask mode                | <code>answer</code> with citations over page and tab digests; streaming                                                       | S    | 7    | E-05       | Not started |
| E-07 | Sequencing and resync   | <code>seq dedupe</code> cache, <code>resync_required</code> , periodic full state, state hash, stale responses dropped        | M    | 8    | E-02       | Not started |
| E-08 | Diff reconstruction     | Server rebuilds current state from base + diffs; property tests against full states                                           | S    | 9    | E-07       | Not started |
| E-14 | Background task support | Task definitions, read-only enforcement, <code>report</code> output                                                           | S    | 9    | E-06       | Not started |
| E-11 | Escalation VLM          | <code>/escalate</code> endpoint, small VLM served, schema-constrained answer merged into state as <code>server_vlm</code>     | M    | 10   | E-03       | Not started |

| ID   | Feature            | Done when                                                                                       | Size | Week          | Deps | Status      |
|------|--------------------|-------------------------------------------------------------------------------------------------|------|---------------|------|-------------|
| E-15 | Deployment         | Docker images, TLS, signed per-install tokens, rate limits, logging that cannot record payloads | M    | 11            | E-02 | Not started |
| E-10 | Multi-action plans | Up to 3 actions per response for predictable form fills, each with its own <code>expect</code>  | S    | 12 (cut line) | E-06 | Not started |

## Lane F — Experience

| ID   | Feature            | Done when                                                                                                                          | Size | Week | Deps       | Status      |
|------|--------------------|------------------------------------------------------------------------------------------------------------------------------------|------|------|------------|-------------|
| F-01 | Side panel UI kit  | Design tokens (light/dark), buttons, action-trace list component, empty and error states; built standalone, mounted by A-10        | S    | 1    | —          | Not started |
| F-02 | Overlay core       | Closed shadow root with <code>all: initial</code> ; annotations anchored to element IDs; follow scroll and resize                  | M    | 2    | B-01       | Not started |
| F-03 | Overlay primitives | Pointer glide, circle, arrow, badge, label, spotlight; 60 fps on mock sites                                                        | M    | 3    | F-02       | Not started |
| F-08 | Companion          | Small character near cursor with idle, listening, thinking, acting, needs-approval states; can be turned off                       | S    | 4    | F-02       | Not started |
| F-15 | Local memory       | Encrypted IndexedDB notes at placeholder level; view and wipe in settings                                                          | S    | 4    | D-05       | Not started |
| F-05 | Agent mode UI      | Live trace in plain language, pause and stop, kill-switch shortcut, target highlight in page                                       | S    | 5    | F-01       | Not started |
| F-04 | Guide mode loop    | Renders <code>point / annotate</code> , waits for the <code>advance_on</code> interaction, advances; handles user going off-script | M    | 6    | F-03, E-12 | Not started |

| ID   | Feature                     | Done when                                                                                                                       | Size | Week          | Deps       | Status      |
|------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------|------|---------------|------------|-------------|
| F-07 | Privacy view                | Shows last outbound payload exactly as the server sees it, including crops; restricted-mode banner with reason                  | S    | 7             | D-11       | Not started |
| F-09 | Voice capture and VAD       | Mic permission page, capture in offscreen doc, Silero VAD segments, push-to-talk wired to A-11                                  | M    | 8             | A-04, A-11 | Not started |
| F-06 | Confirmations and approvals | High-risk and destination-binding prompts in the side panel only, naming field and site; approve, edit or decline               | S    | 9             | D-09       | Not started |
| F-12 | Ask mode UI                 | Streams answers, resolves placeholders locally for display, citations focus the tab and highlight the element                   | S    | 9             | E-13       | Not started |
| F-10 | Local speech-to-text        | Whisper-tiny or Moonshine-class via Transformers.js on WebGPU; lazy voice pack; English + Hindi; transcripts sent to PII engine | M    | 10            | F-09, C-02 | Not started |
| F-13 | Tab orchestrator            | Tab-grant picker, per-tab sanitized digests (300–800 tokens) rebuilt on change, max 8 tabs                                      | M    | 11            | A-05, D-10 | Not started |
| F-11 | Local text-to-speech        | speechSynthesis limited to local voices; streams say ; barge-in stops speech                                                    | S    | 12            | F-10       | Not started |
| F-14 | Background tasks            | Schedule UI, alarms , minimised window (Firefox: background tab), read-only default, approval queue, notifications              | M    | 13 (cut line) | E-14, A-05 | Not started |